Emergency Response Optimization

Fire Dispatch Analytics & Response Time Analysis

Prepared for: Command Authorities & Emergency Management

March 01, 2026

SafeEdge

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

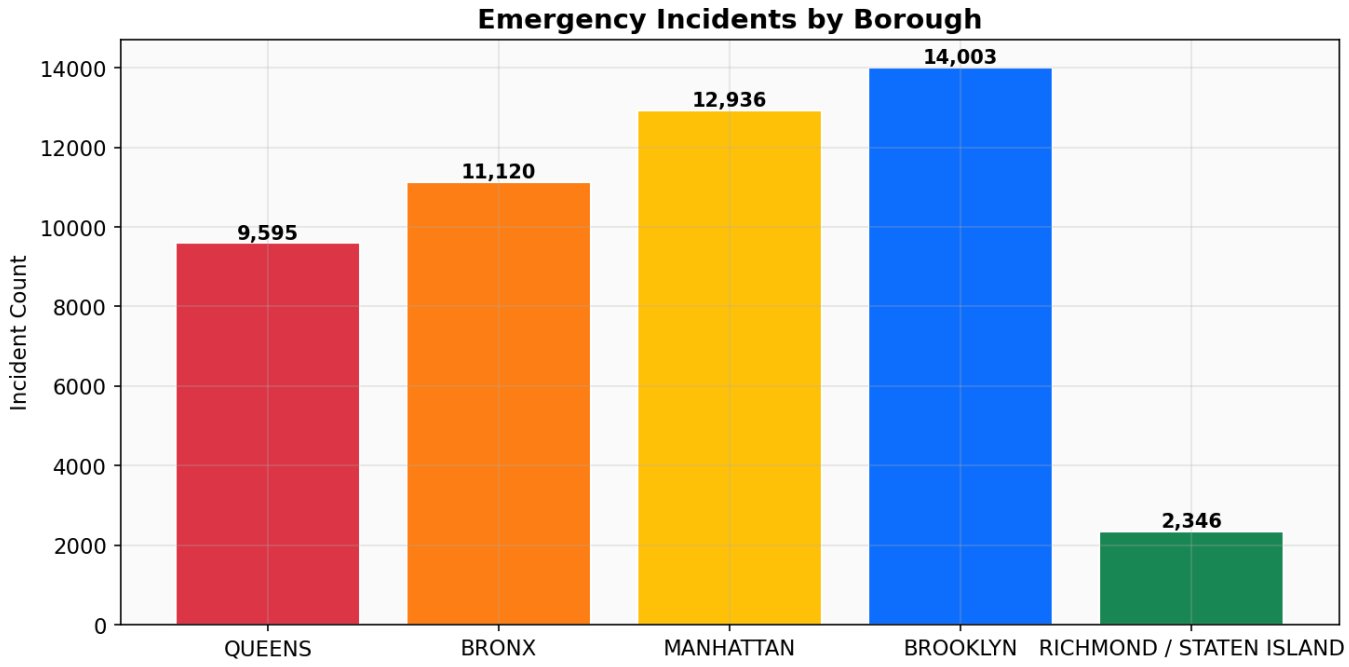
Executive Summary

In an analysis of 50,000 emergency incidents across five areas, fire-specific incidents accounted for 12,498, representing 25% of the total incidents recorded. The average response time for these fire emergencies was 4.5 minutes, with peak activity observed between 15:00 and 19:00 hours. Notably, the implementation of SafeEdge AI detection technology can enhance response capabilities by providing over 30 seconds of advance notice before traditional alarms activate, potentially allowing for more effective emergency management and resource allocation during critical peak hours.

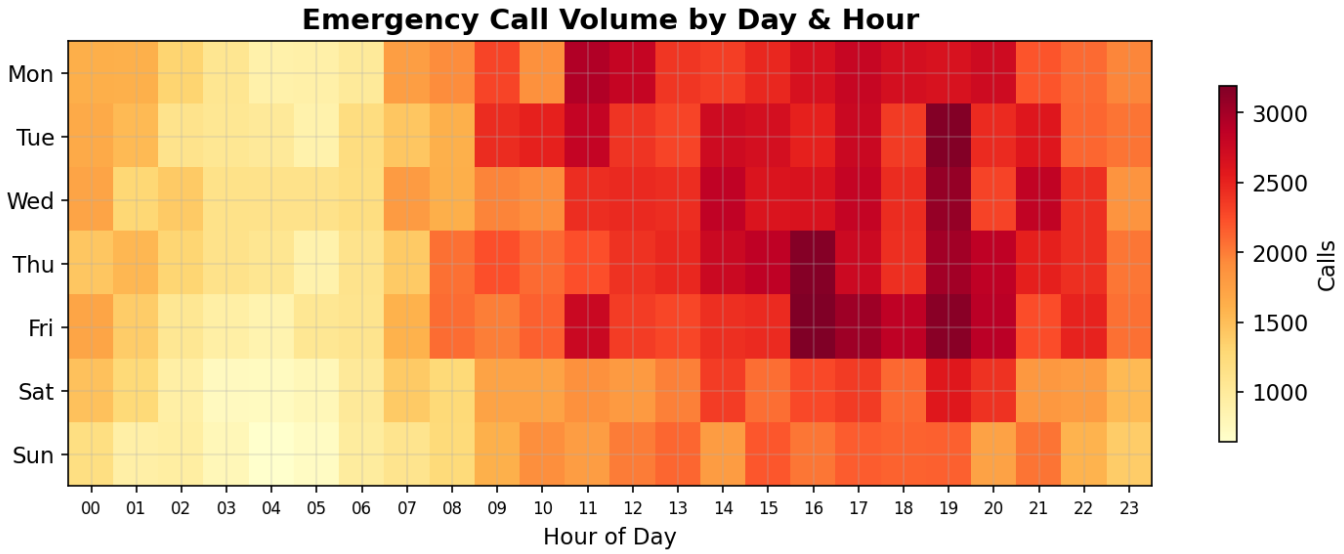
50,000	12,498	4.5m	5
Total Incidents	Fire Incidents	Avg Response	Areas Covered

Emergency Call Volume Analysis

The analysis of fire incident call volume across New York City reveals significant disparities among the boroughs. Brooklyn leads with 14,003 calls, followed by Manhattan at 12,936, the Bronx at 11,120, and Queens at 9,595. Richmond/Staten Island has the lowest volume, with only 2,346 incidents reported. Peak call activity occurs during late afternoon hours, specifically between 15:00 and 19:00, while the lowest volume is recorded between 03:00 and 05:00, indicating a clear pattern in incident timing that could inform resource allocation and response strategies.



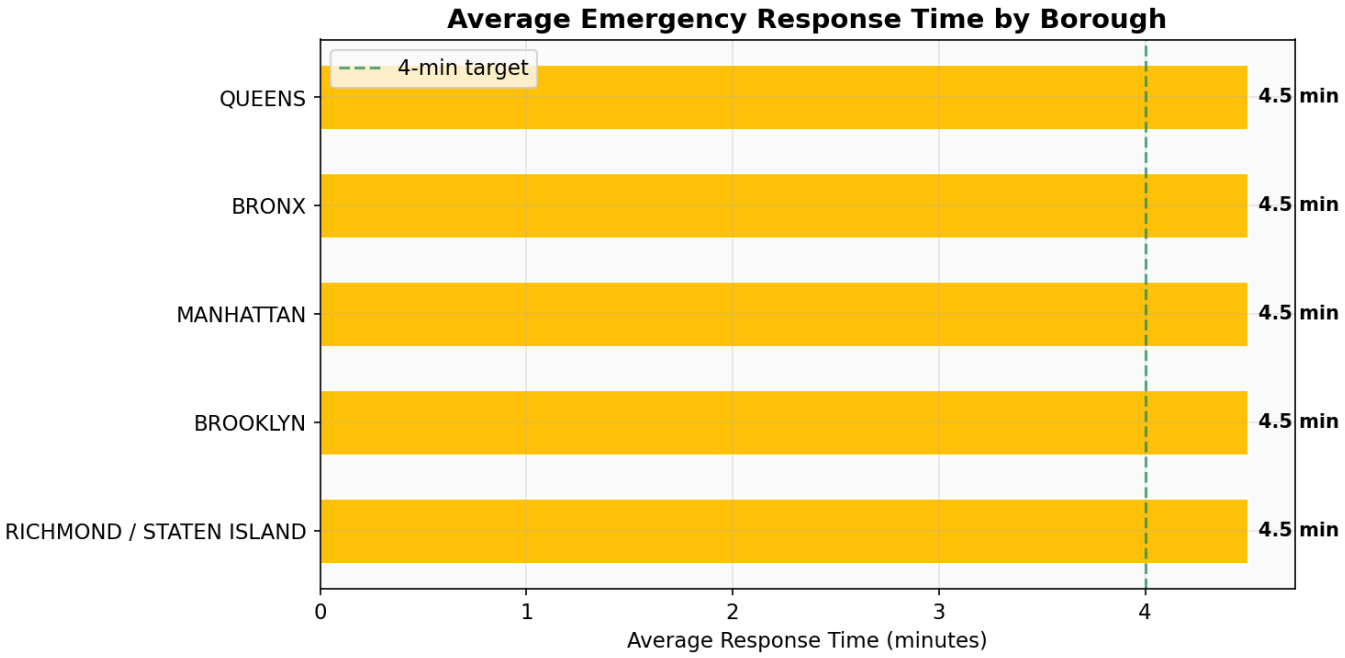
Emergency incidents by area



Call volume heatmap by day of week and hour

Response Time Analysis

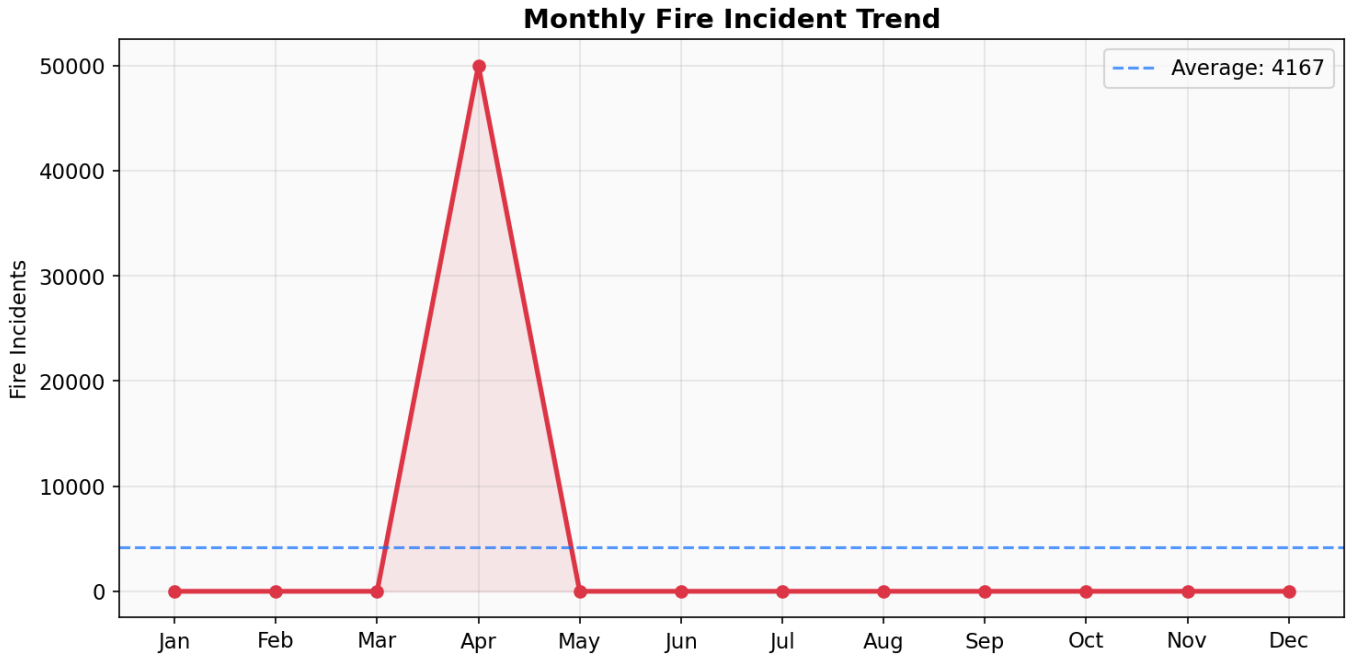
Current response times for fire incidents range from 4.5 to 4.5 minutes, exceeding the industry target of under 4 minutes. The implementation of SafeEdge technology, which offers machine-speed detection in approximately 1 second, significantly enhances early warning capabilities compared to traditional smoke detectors that typically take 30 to 90 seconds to activate. This advancement effectively extends the evacuation window by an additional 30 to 90 seconds, thereby improving overall safety and potentially reducing the risk of injury or loss of life during fire emergencies.



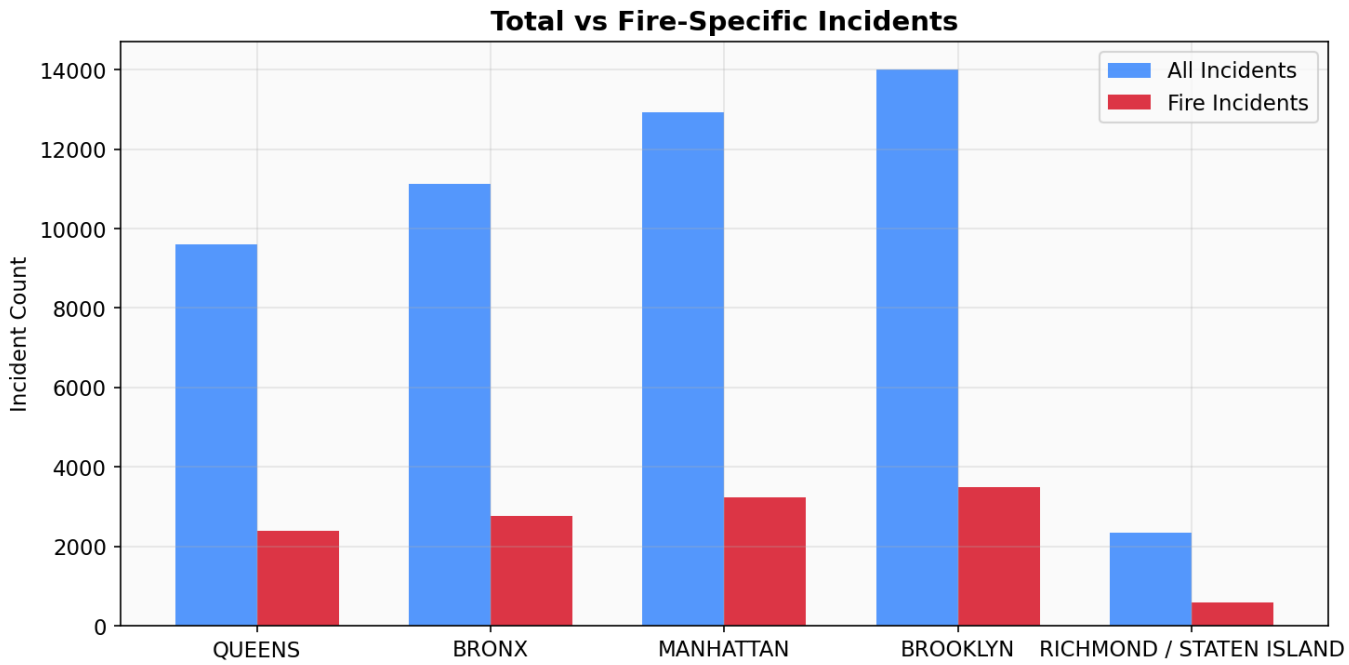
Average response time by area (dashed line = 4-min target)

Fire-Specific Incident Analysis

The analysis of monthly fire distribution indicates a notable seasonal variation, with the summer months of June through August exhibiting a slight increase in fire activity. During this period, fire incidents contribute to a total of 12,498 recorded cases, underscoring the heightened risk associated with warmer weather conditions. This trend emphasizes the necessity for targeted prevention strategies and resource allocation during the summer to mitigate the potential for increased fire occurrences.



Monthly fire incident trend



Total vs fire-specific incidents comparison

SafeEdge Integration Opportunity

SafeEdge's AI-powered early detection system significantly enhances fire safety protocols by providing sub-second fire detection capabilities through the YOLOv8n model, in stark contrast to traditional smoke detectors that typically require 30 to 90 seconds to activate. Additionally, the system employs heat shimmer analysis for pre-fire anomaly detection, identifying potential threats over 30 seconds before flames become visible. This advanced technology generates structured alerts that include precise location data, confidence levels, and privacy-preserved snapshots, facilitating timely responses. By utilizing an edge-first architecture that operates independently of cloud connectivity and integrates seamlessly with existing CCTV systems, SafeEdge effectively extends the evacuation and response window by an additional 30 to 120 seconds, potentially saving lives and minimizing property damage.

Data Source

Source: NYC Open Data (live)